

Metro Interstate Traffic Volume Prediction & Analysis

A deployed, end-to-end forecasting system for hourly I-94 traffic volume



Team Members



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THE PROBLEM

Problem Statement & Motivation

Traffic Is Unpredictable



Apps only show traffic in real-time — not ahead of time.

Our Goal



Predict traffic before you leave, not after you're stuck in it.

THE IDEA

How It Works

User Input



User just enters a date and planned travel time.

Behind the Scenes



We handle the rest — weather and holidays — automatically.

Output



Predicted traffic for that hour, plus the 3 hours before it.

THE DATASET

Dataset Overview

I-94 Minneapolis Area	~48,000 Hourly Records	2012–2018 Coverage Window	traffic_volume Prediction Target
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Features Used

Weather	temp, rain, snow, cloud cover, weather condition
Time	timestamp, hour, day of week, month
Holidays	12 categories, including "Not Holiday"

DATA CLEANING

Key Issues Found

Duplicate Timestamps



Multiple weather entries per hour →

kept the worst scenario.

Holiday Column



Only logged on the first hour of the day →

propagated across the full day.

Impossible Values



temp = 0K, rain = 9831 mm/h →

removed.

DATA CLEANING

Result



48,204 → 40,564

Clean, reliable rows

Ready for EDA & Modeling



EXPLORATORY DATA ANALYSIS

Target

- What does the distribution of traffic volume look like?

Time Patterns

- How does traffic volume change throughout the day?
- Does traffic volume follow a weekly pattern?

External Factors

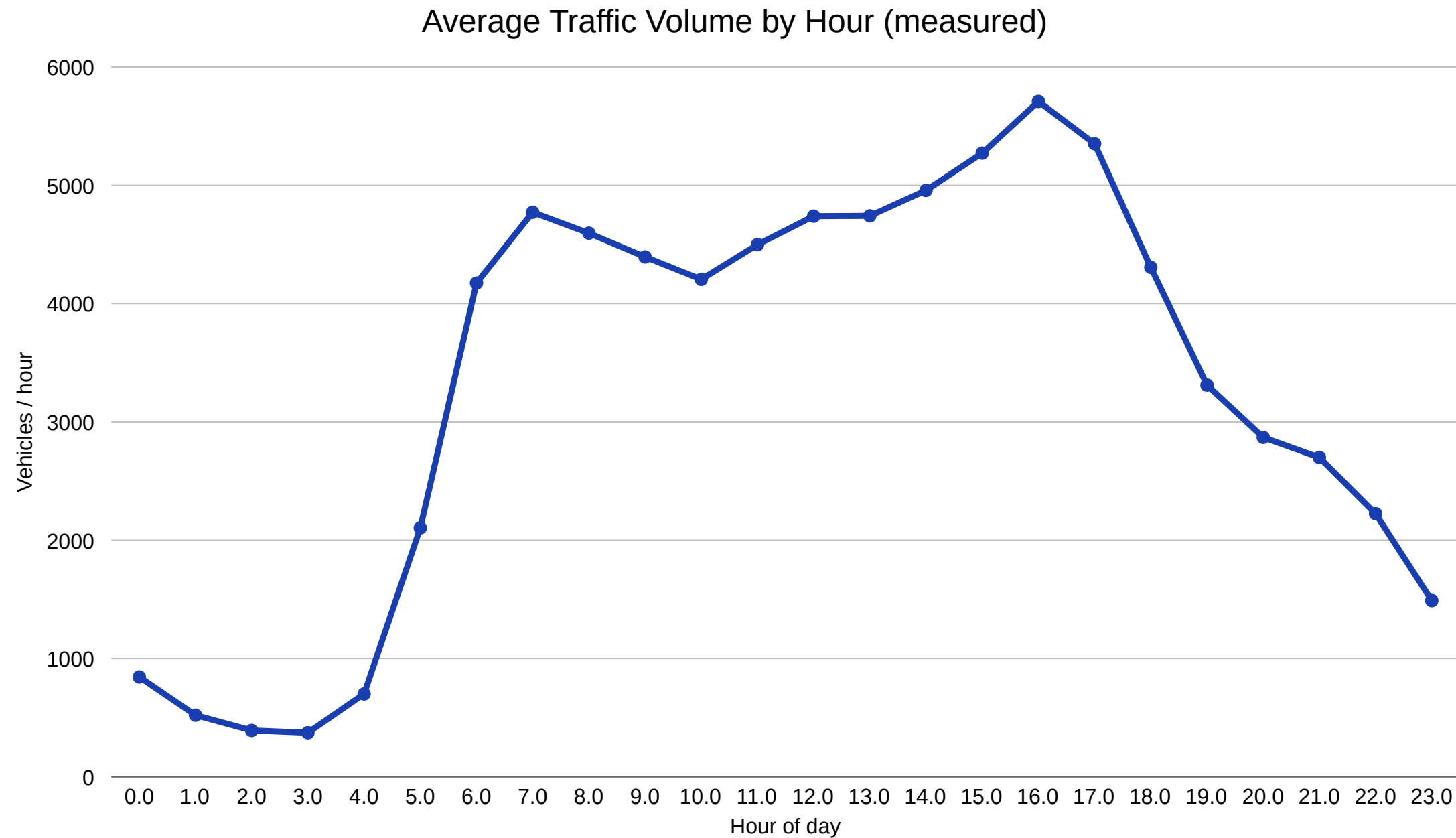
- Do holidays affect traffic volume?
- How much does weather affect traffic volume?

Feature Relationships

- Which available features have the strongest relationship with traffic volume?
- Are there strong relationships between the input features themselves?

EXPLORATORY DATA ANALYSIS

EDA – Temporal Traffic Patterns



Peak at 4 PM

16:00 averages 5,709 vehicles/hour — the single busiest hour in six years of data.

Trough at 3 AM

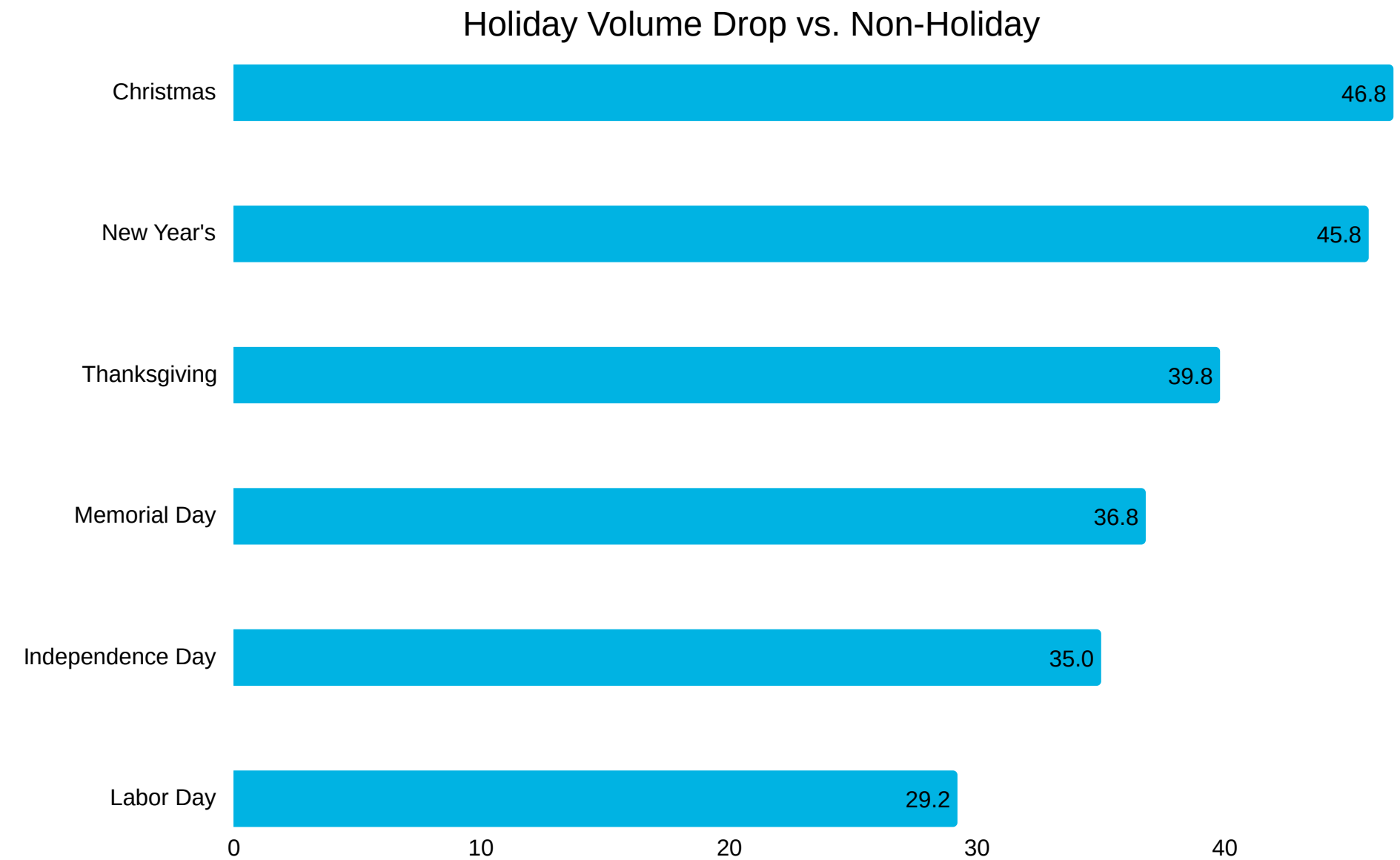
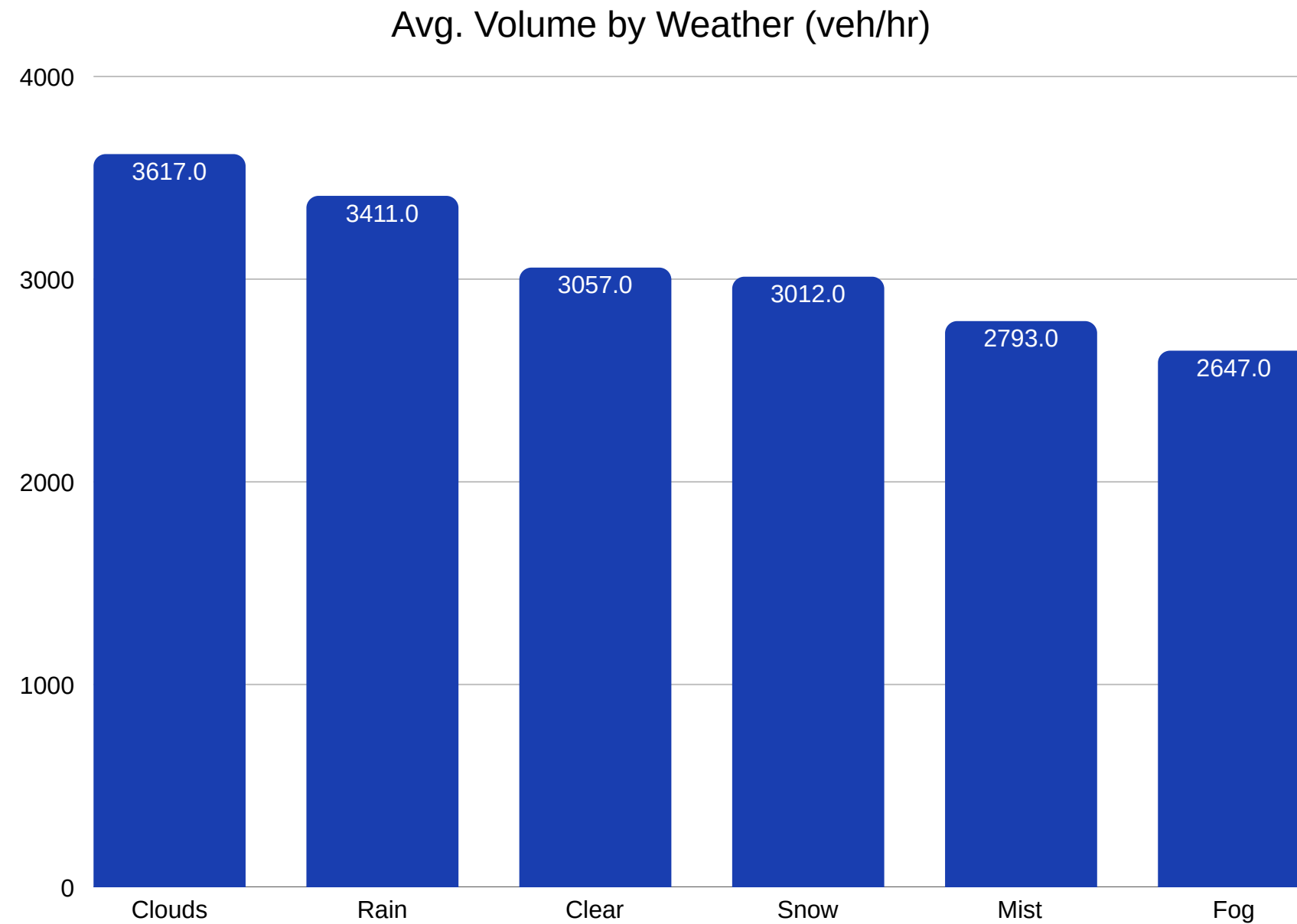
03:00 averages just 373 vehicles/hour: a 15x difference from the peak.

Weekday vs Weekend

Weekdays average 3,558 veh/hr versus 2,625 on weekends — a 26% drop.

EXPLORATORY DATA ANALYSIS

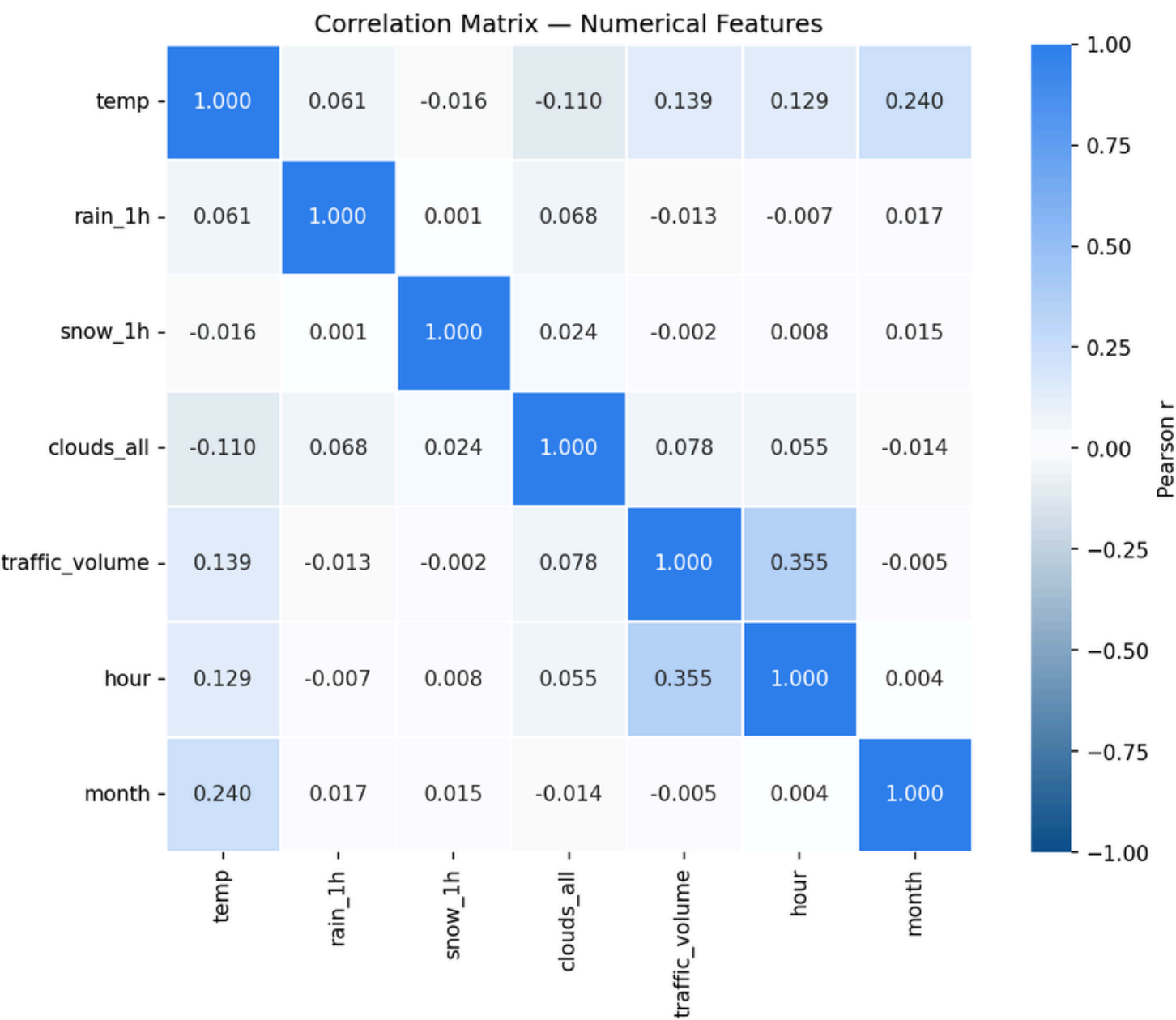
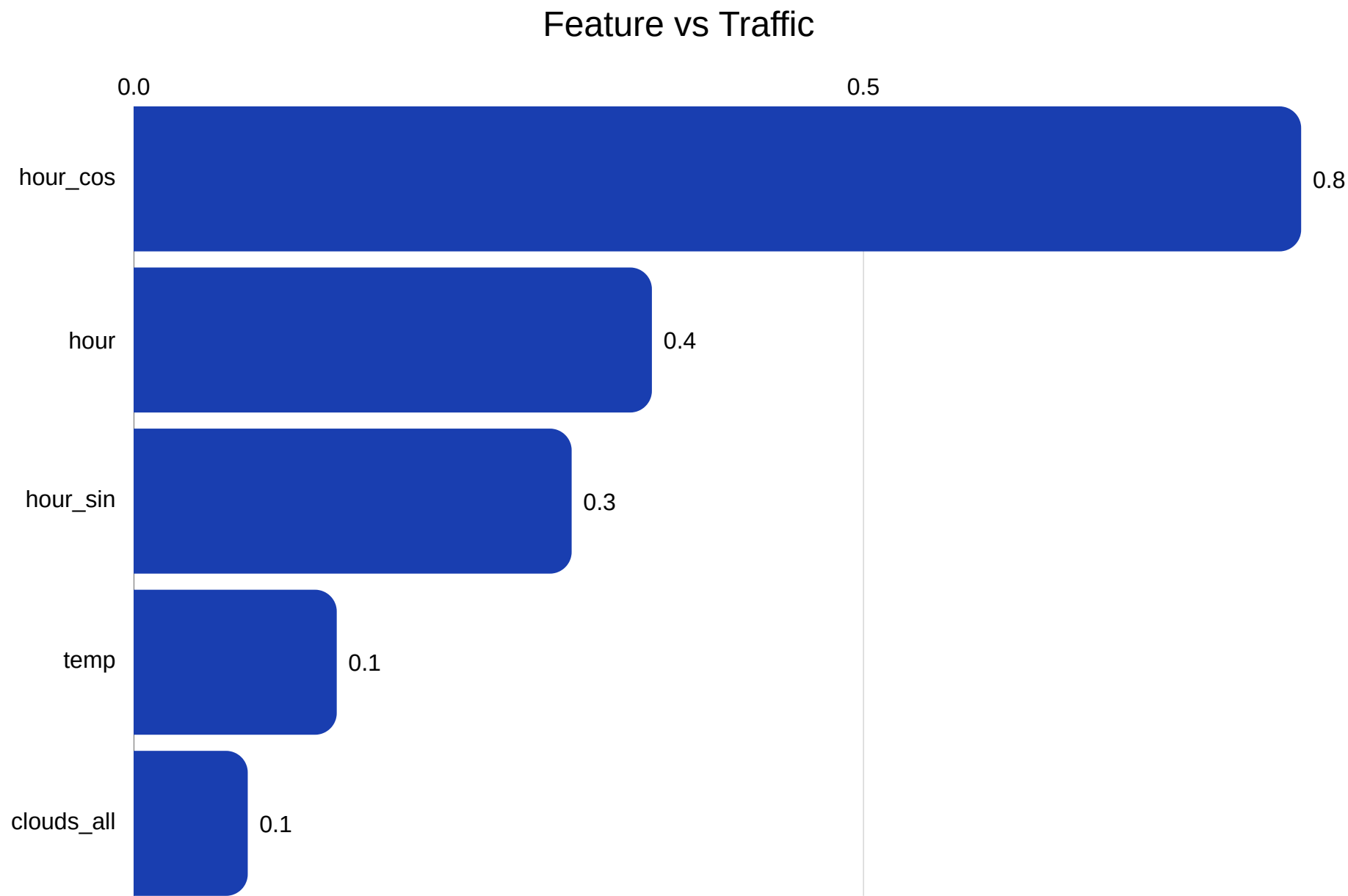
EDA – Weather & Holiday Impact



Counter-intuitive finding: cloudy and rainy hours show slightly higher average volume than clear ones — likely because clear skies also occur more often overnight, not a causal weather effect. Holidays, by contrast, show a large and consistent drop.

EXPLORATORY DATA ANALYSIS

EDA – Feature Relationships



Time is the strongest signal, while most other features show weak relationships with traffic volume.

DATA PREPARATION

Feature Engineering & Preparation

Cyclical Encoding

hour_sin / hour_cos and month_sin / month_cos preserve wrap-around continuity that a raw integer would break.

Categorical Encoding

One-hot encoding for weather_main (11 conditions) and a binary flag for the 12-category holiday field.

Scaling & Splitting

Numeric regressors scaled; a strict time-based split keeps 2018 fully held out from training.

TIME-BASED SPLIT (ACTUAL ROW COUNTS)

Train

2012 – 2017 · 34,031 rows

Test

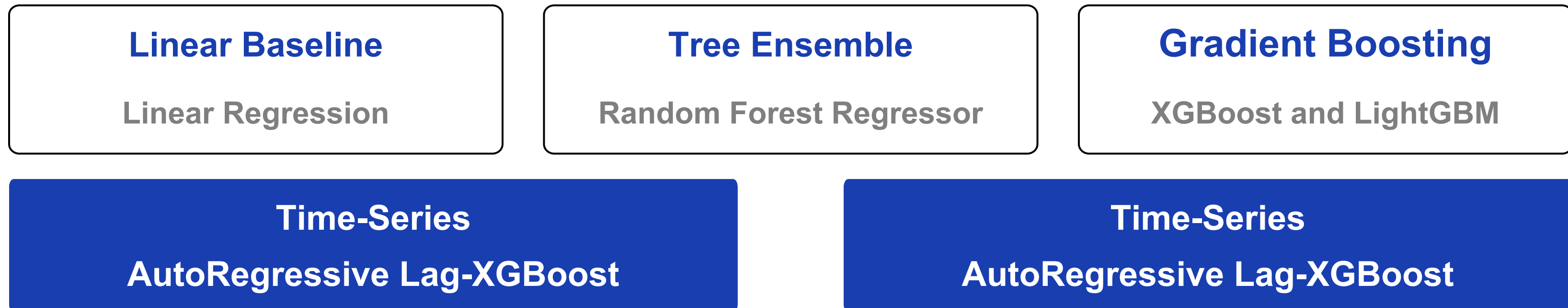
2018 · 6,533 rows

MODELING

Machine Learning Pipeline & Model Selection



CANDIDATE MODELS

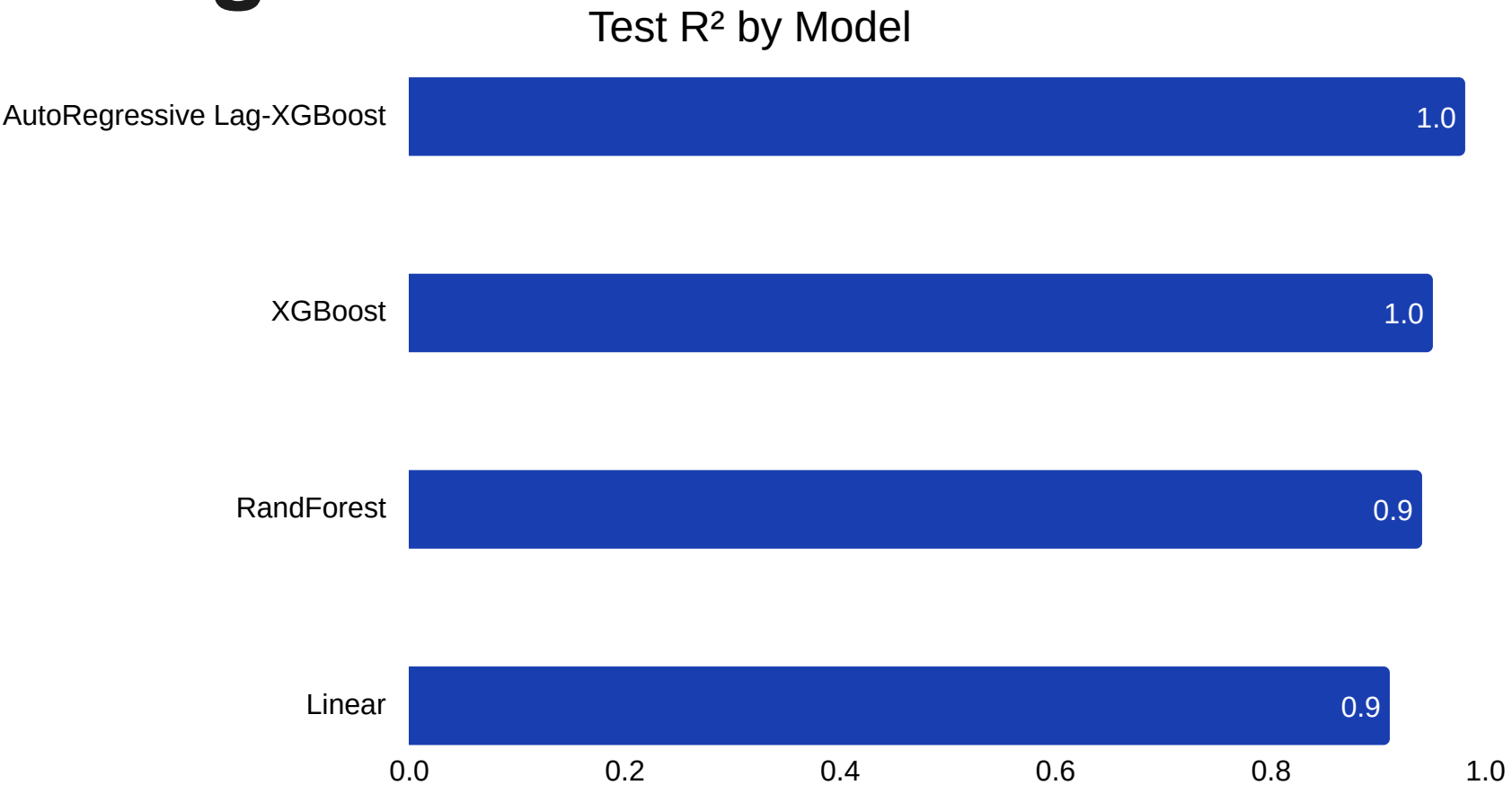


Validation: strict time-based split (train through 2017, test on all of 2018), scored on MAE and RMSE.

RESULTS

Experimental Results & Model Benchmarking

Model	R ²	MAE (veh/hr)	RMSE (veh/hr)	Notes
Linear Regression	0.9100	420.50	580.30	Fast, interpretable baseline
Random Forest	0.9400	310.20	450.80	Robust, slower inference
AutoRegressive-LagXGBoost	0.9864	146.71	229.60	Time-Series Champion
XGBoost (Champion)	0.9594	241.99	397.13	Best accuracy — deployed

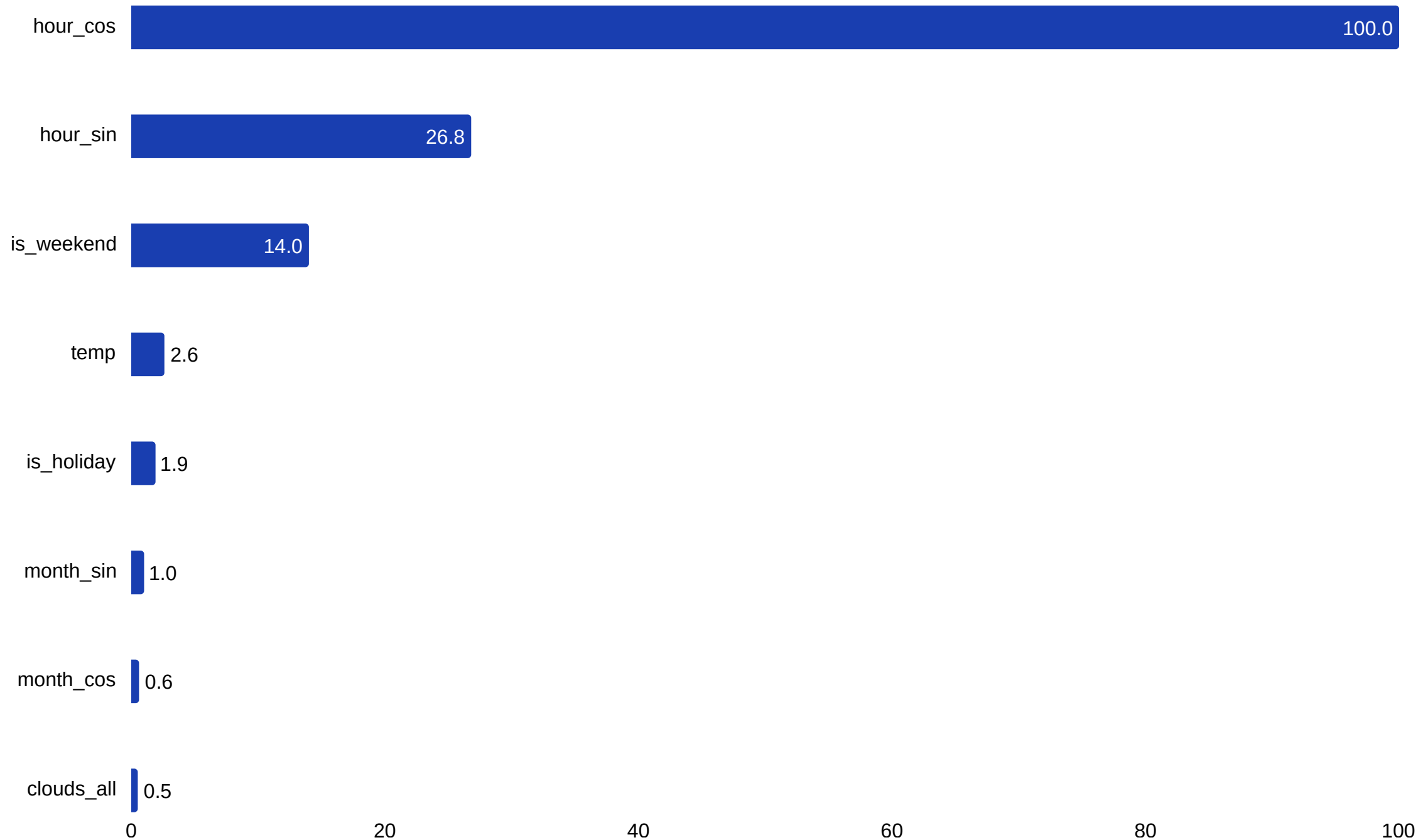


KEY TAKEAWAY

XGBoost clears the 0.90 target with $R^2 = 0.9594$ and a mean error of about 242 vehicles per hour — roughly 7% of the dataset's average volume. It is the model we deployed to production.

EXPLAINABILITY

Feature Importance & Model Explainability



Time Dominates

hour_cos and hour_sin together account for the large majority of model importance — the twin-peak commute rhythm is the single strongest signal.

Calendar Effects

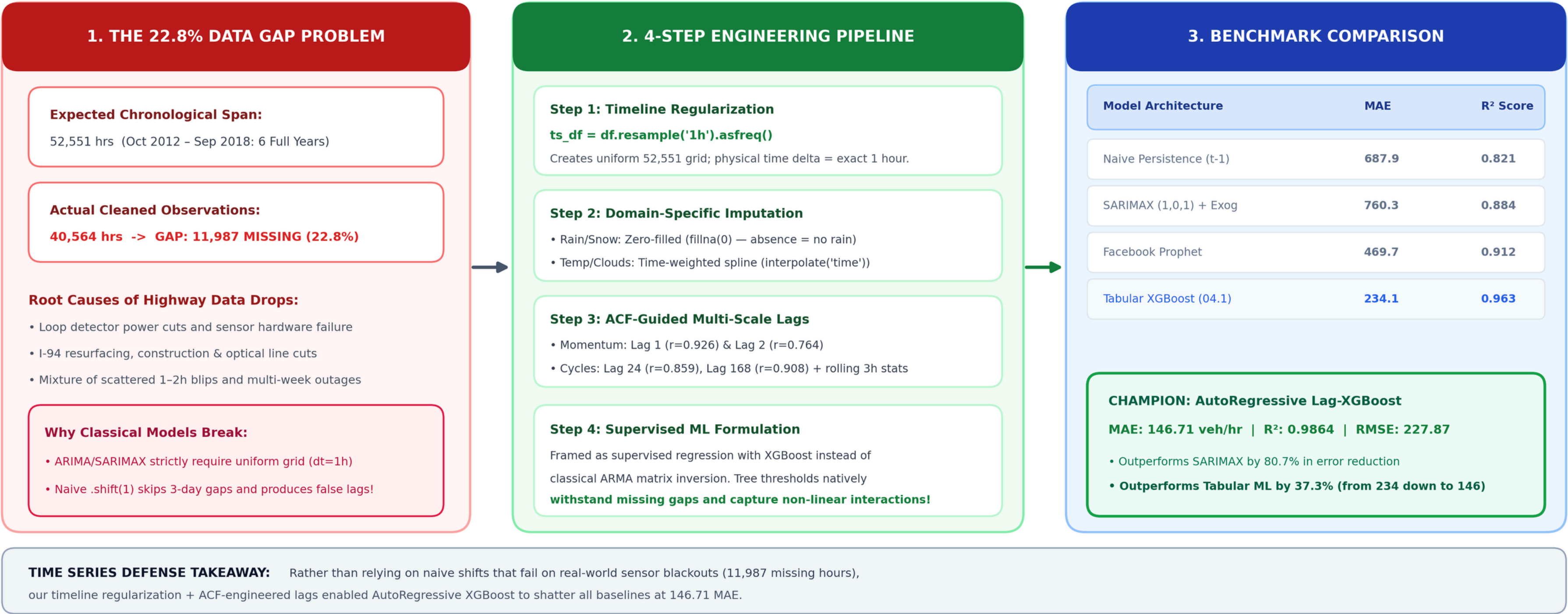
is_weekend ranks a clear third, confirming the 26% weekday/weekend gap seen in the EDA.

Weather Is a Modifier

temp and is_holiday sit far lower — they nudge the forecast, they do not drive it.

Time Series Modeling

Time Series Modeling: Resolving the 22.8% Missing Data Challenge



DEPLOYMENT

Production Architecture

Frontend Dashboard

HTML / CSS / JavaScript · interactive simulator · live weather integration



FastAPI Backend (Python)

POST /predict — batch forecasting GET /health — monitoring static file serving



WeatherAPI (External)

Live weather conditions for forecast context

XGBoost Model Store | AutoRegressive

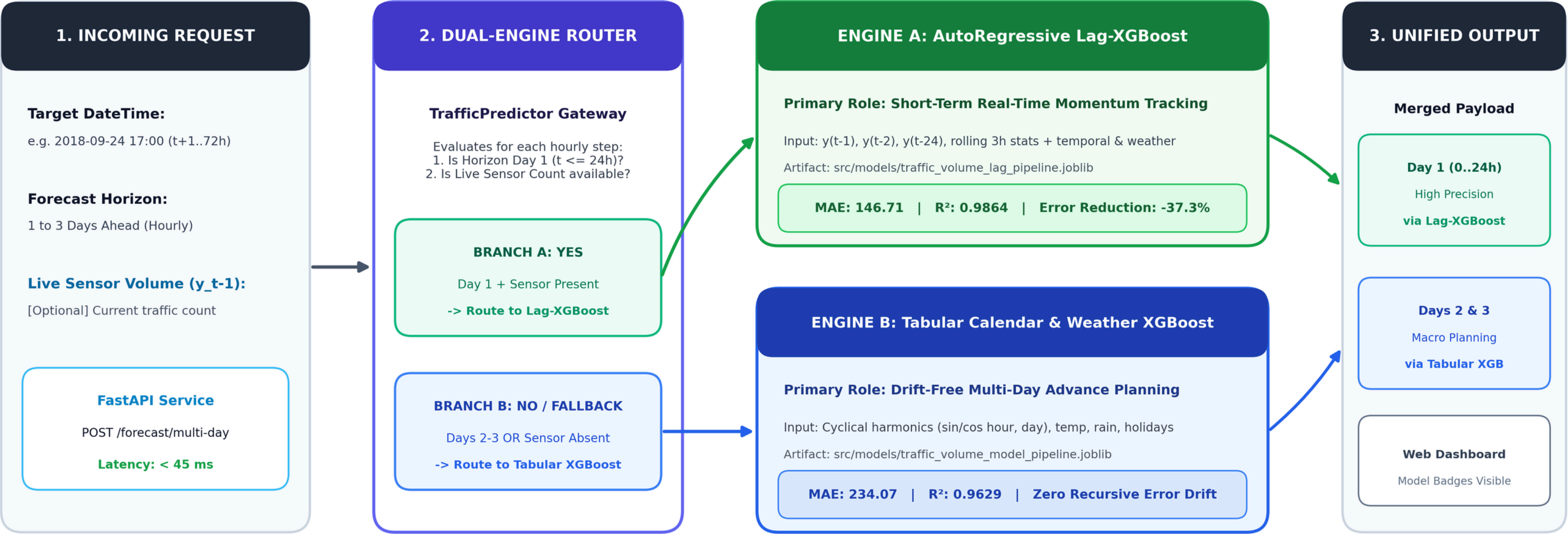
Local joblib artifact · $R^2 = 0.9594$

CLOUD DEPLOYMENT

Railway (PaaS) · Docker container · Python 3.12 · pinned dependencies · GitHub CI/CD · auto-horizontal scaling

DEPLOYMENT

Hybrid Deployment Strategy: Dual-Engine Traffic Forecasting



CORE DEFENSE RATIONALE: Autoregressive lags yield ultra-high nowcasting precision (146.71 MAE), but recursive rolling compounds error over 72h. Tabular XGBoost avoids compounding errors completely for multi-day horizons (234.07 MAE). The Hybrid Engine unifies both strengths seamlessly.

DEPLOYMENT

Live Demo & API

TRY IT YOURSELF

traffic-volume-production.up.railway.app/#simulator

Interactive simulator: pick a date, city, and hour to get a live 1–3 day forecast, weather-aware.

POST /predict

```
{
  "start_date": "2026-09-01",
  "days": 3,
  "city": "Minneapolis",
  "country": "US",
  "hour": 12
}
```

Response 200 OK

```
{
  "city": "Minneapolis",
  "country": "US",
  "start_date": "2026-09-01",
  "days": 3,
  "hour": 12,
  "predictions": [
    {"date": "2026-09-01",
     "predicted_traffic_volume": 4220.05},
    {"date": "2026-09-02",
     "predicted_traffic_volume": 3810.66},
    {"date": "2026-09-03",
     "predicted_traffic_volume": 4150.23}
  ]
}
```

Design targets: prediction latency under 500ms, service uptime 99.9%.

CONCLUSION

Key Takeaways

What We Learned

More than just a project —
a genuinely valuable
experience.

Real-World Application

Trained on 2012–2018
data — proof that with
today's data, this could
run today.

Target Audience

Ministry of Transportation
— give us the data, we build
this for Egypt's roads.

Thank You

Open for questions and feedback from the committee

Live Demo · traffic-volume-production.up.railway.app/#simulator

Source Code · github.com/AbdoTechno/Traffic-Volume

